Name: _____

WORKSHEET

Application of optimisation

1 A bottle rocket is launched from ground level. The height h in metres after t seconds is given by $h(t) = 42t - 5t^2$. Find:

a the time it takes the rocket to reach maximum height.

b the maximum height reached by the bottle rocket.

2 The displacement of a particle from a reference position is given by $x(t) = t^3 - 3t + 5$. What is the closest that the particle gets to the reference position for $t \geq 0$?

3 P is a movable point on the line $y = 7 - x$. Find the coordinates of P when it is closest to the origin.

4 Find the point (x, y) on the line $y = -2x - 5$ which is a minimum distance to $(3, 7)$.

5 A point X on the curve $y = x^4 + x^3 - 4x^2 - 5x + 5$ is vertically aligned with a point Y on the curve $y = x^3 - 5x - 3$.

a Show that the distance between X and Y is given by $d = x^4 - 4x^2 + 8$.

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b Find the minimum distance between X and Y.

c Find the maximum distance between X and Y on the interval $-2 < x < 2$.

6 Find the maximum volume of a cylinder if the sum of the radius and height is equal to 30 cm.

7 A 120 m length of fence is to be made into the boundary of a rectangular paddock.
What is the maximum area it can enclose?

8 A piece of wire, 24 cm long, is bent to form the two shorter sides of a right-angled triangle.
What is the maximum area of the triangle?

9 A piece of cardboard is to be made in to an open box of maximum volume by removing a square of side x cm from each corner and folding up the sides. If the cardboard is 20 cm by 16 cm, determine x and find the maximum volume of the open box.

10 Find two numbers that have a sum of 38 and a maximum product.

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11 Find two numbers whose difference is 32 and whose product is a minimum.

12 The cost to manufacture x items per day is given by: $c(x) = 20x^3 + 7.5x^2 - 6150x + 45\,000$.

a Find the number of items to produce at a minimum cost

b What is the minimum cost?

13 A company that manufactures lasers receives \$25 in revenue per laser. The cost of making x lasers is given by $c(x) = x^3 - 35x^2 - 795x + 70$.

a Write an equation for the profit made from selling x lasers

b How many lasers should be made and sold in order to maximise profit?

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Answers

1 **a** $t = 4.2s$

b $h_{max} = 88.2$ m

2 $x(1) = 3$ units

3 $P = \left(\frac{7}{2}, \frac{7}{2}\right)$

4 $(-4.2, 3.4)$

5 **a** $d = x^4 - 4x^2 + 8$

b $d_{min} = 4$ units

c $d_{max} = 8$ units

6 $V_{max} \approx 12\,566$ cm^3

7 $A_{max} = 900$ m^3

8 $A_{max} = 72$ cm^3

9 $V_{max} \approx 420$ cm^3

10 19 and 19

11 16 and -16

12 **a** 10

b \$4250

13 **a** $p(x) = -x^3 + 35x^2 + 820x - 70$

b 32

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